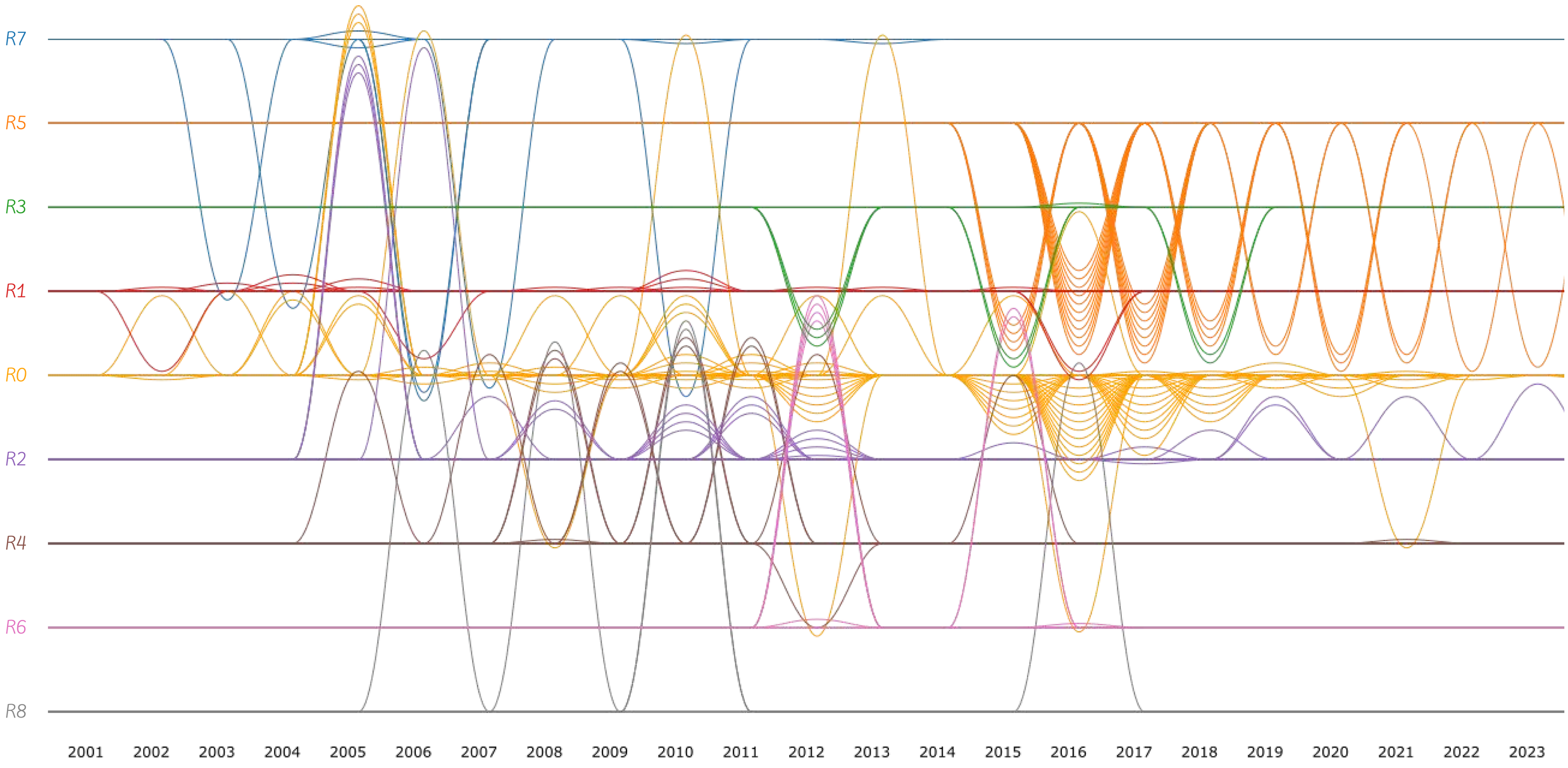
ThreadWave

Visualizing Co-authorship as Temporal Threads

We present a timeline-based visualization for exploring temporal collaboration patterns among academic authors. Inspired by the metaphors of weaving threads and ripple waves, our technique represents each author as a colored baseline and uses curved connectors to show co-authorship events over time.

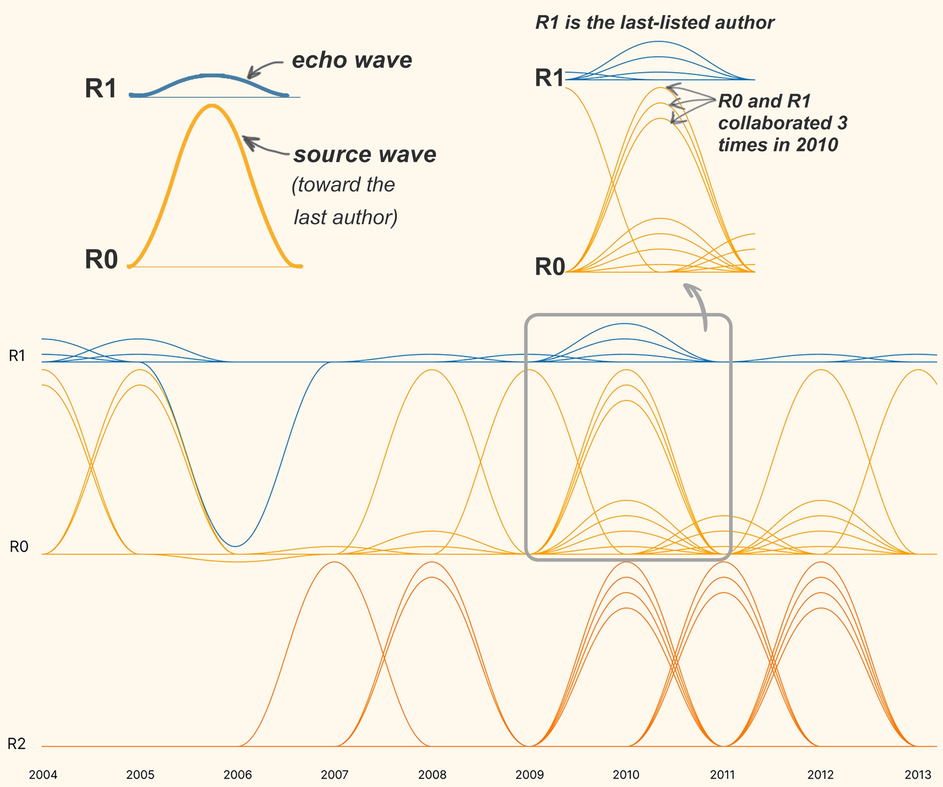


ThreadWave visualization depicting the temporal and relational dynamics of researcher R0's collaborations. The X-axis represents time, while the Y-axis lists nine researchers (R0–R8), each assigned a distinct color. Curved lines (i.e., waves) illustrate co-authorship events, revealing when and with whom R0 collaborated over their research journey.

Motivation

Academic collaboration is often visualized with network graphs that show who collaborated with whom. Yet these approaches overlook the temporal dimension: when collaborations began, how they evolved, and when they faded. Existing tools such as VOSviewer, Gephi, or Cytoscape emphasize connectivity, but they flatten time and disrupt the continuity of collaboration stories.

Visual encoding in ThreadWave



- Each researcher = a colored baseline (line over time).
- Collaboration = Bézier "wave" curve linking authors to the last author.
- Overlapping ripples = multiple collaborations in the same year.
- Inspired by storyline visualization and narrative metaphors.

Highlight in Largest Area

- **Continuity & Storytelling:** Shows collaborations as evolving journeys.
- **Clarity:** Quickly see when, with whom, and how often collaborations occurred.
- **Pattern Discovery:** Identify frequent collaborators and shifts in research partnerships.
- **Narrative Metaphor:** Collaboration visualized as threads & ripples → intuitive, human-readable patterns.

Limitations & Future Work

Limitations: Currently designed for small-to-medium collaboration groups.

Future work:

- Scale to larger networks.
- Integrate metadata (topics, venues).
- Add interactive filtering/exploration.
- Apply to other domains (team dynamics, creative work).

Implications & Take-aways

Collaboration is more than a static connection—it is a narrative unfolding across time. ThreadWave transforms co-authorship data into a temporal story, enabling audiences to see influence, continuity, and change. This perspective invites richer academic analysis and fosters new opportunities for collaboration.

Acknowledgement

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Ruishan Wu

Zezhong Wang

Charles Perin

Sheelagh Carpendale

INNOVIS
ixlab
SFU
SIMON FRASER UNIVERSITY